

Lab Report: Processor Design and Implementation

1. Introduction

In this lab, we designed and implemented a simple processor by developing individual modules and integrating them into a complete system. The key components include the Register File, Arithmetic Logic Unit (ALU), Instruction Memory, Immediate Generator, and Control Unit.

Each module was first designed and verified independently. In Lab 11, all modules were integrated to form a functional processor capable of executing assembly instructions. The objective of this lab is to run assembly programs on the processor, extend its functionality by adding a new instruction, and verify the updated design.

2. Task A: Running Assembly Program

Steps Taken:

1. Developed and verified all required RTL modules including ALU, Register File, Instruction Memory, and Immediate Generator.
2. Integrated all modules into a top-level processor design.
3. Created a testbench to simulate the processor behavior.
4. Loaded the assembly program into the instruction memory.
5. Applied appropriate constraints for synthesis and implementation.
6. Ran simulation to verify correct execution of instructions.
7. Observed waveform outputs to confirm expected results.

The processor successfully executed the assembly instructions, and the outputs matched the expected behavior.

3. Task B: New Instruction Implementation

Instruction Description:

A new instruction, [Instruction Name], was implemented. This instruction performs [brief description of operation].

Datapath Modifications:

- Added necessary logic to support the new instruction.
- Updated multiplexers and data routing paths where required.
- Ensured compatibility with existing datapath components.

Control Path Changes:

- Modified the control unit to generate appropriate control signals for the new instruction.
- Added new control signal conditions based on opcode decoding.
- Updated instruction decoding logic.

Verification:

Updated the testbench to include test cases for the new instruction and verified functionality through simulation.

4. Task C: Program Execution with Updated Design

Steps Taken:

8. Loaded the updated assembly program into instruction memory.
9. Used the modified processor design from Task B.
10. Applied constraints and ran simulation.
11. Verified execution through waveform analysis.

The processor successfully executed the program, including the newly added instruction, confirming correct integration.

5. Conclusion

In this lab, we successfully designed and implemented a processor by integrating multiple modules. We executed assembly programs and extended the processor by adding a new instruction. The modifications to the datapath and control unit were verified through simulation. Overall, the lab provided a clear understanding of processor design, instruction execution, and system-level integration.